Integrating multiomics data to predict clinical outcomes in high-grade serous ovarian cancer

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October 15, 2026

Abstract

High-grade serous ovarian cancer (HGSC) is the most lethal gynecological cancer and remains a major clinical challenge due to its heterogeneous disease progression and limited treatment response. The tumor microenvironment (TME) has been shown to influence disease outcome and therapeutic response. However, complex interactions between molecular characteristics, immune context, and clinical outcomes are not fully understood. This project aims to integrate clinical, genomic, and spatial omics to identify biological features associated with clinical outcomes in HGSC patients. The dataset consists of clinical information, whole exome sequencing data, and spatial profiling data generated using tCycIF from tissue samples. Due to the high-dimensional low-sample-size nature of multi-omics data, hybrid feature selection is applied to reduce dimensionality and improve model interpretability. From filter methods, correlation-based approaches and mRMR algorithm are used, and embedded methods are applied in model training to select feature sets for each model. Simple and explainable models are prioritized to reduce the risk of overfitting and to allow biological interpretation of the identified features. Linear regression and accelerated failure time (AFT) survival models were used to predict progression-free survival. Predictive performance was modest. Linear regression had a mean absolute error of 174 days. The AFT models ranked patients better than random with mean C-indices of $0.67 - 0.70$, and the log-logistic model had the lowest mean MAE, 153 days. PTEN and KRAS copy number and weighted LOH of chromosome 10 were the most consistently selected features, together with spatial features describing lympho- and myelonets and immune communities. These results should be considered as exploratory and hypothesis-generating candidates.

1 Introduction

High-grade serous ovarian cancer (HGSC) is the most lethal gynecological cancer and remains a clinical challenge due to its heterogeneous patient outcomes [5]. The disease is often diagnosed late, when the tumor already has metastasized. One of the characteristics of HGSC is its rich immune microenvironment, which is why immunotherapies are explored as a promising treatment. Despite recent advances in treatment strategies, patients still face poor survival [2].

One of the key characteristics of HGSC is its complex tumor microenvironment (TME), which includes immune cell populations and interactions between tumor and immune cells. Interactions between immune cells and tumor tissue can influence disease progression and treatment response [2, 6]. Recent studies show, that the composition and spatial organization and interactions of immune cells in TME play a major role in influencing the clinical outcome of HGSC patients [3], which is why better understanding of the biological mechanisms is needed.

Integrating multiple layers of biological data provides an opportunity to identify complex relationships that cannot be captured using single omics data. Combining clinical information with genomic and spatial omics, enables the combination of molecular information and spatial context. The data used in this project come from tissue samples that have been analyzed using multiplex fluorescence imaging (tCycIF) and GeoMx spatial transcriptomics. These methods allow the characterization of immune cell localization patterns and gene expression profiles within the TME. The analysis focuses mainly on CD4+ T cells, CD8+ T cells, macrophages and dendritic cells. These immune cells play a critical role in the progression of the disease and the response to treatment.

The project integrates clinical information with genomic and spatial omics data through exploratory data analysis and feature selection to predict patient outcomes with machine learning models. The aim is to identify combinations of features associated with clinical outcomes of HGSC with different genetic backgrounds. The aim of this project is to implement simple, explainable machine learning models to maintain biological interpretability and reproducibility of the results.

Section 2 describes the dataset, including its characteristics and preprocessing steps. Section 3 presents the methods used in the study, while Section 4 presents the results. Section 5 discusses the results and key findings, and Section 6 concludes the report.

2 Data

This project focuses on multi-omics data, more specifically in combining clinical and genomic information of the patients and spatial information of different immune cells in the samples. The data used in this project consist of 42 different patients and a total of 58 Formalin-Fixed Paraffin Embedded (FFPE) HGSC tissue samples, with some patients contributing multiple samples. Whole exome sequencing (WES) and comprehensive clinical data has been collected as part of Oncosys-OVA clinical study. WES data enables the exploration of patients' immuno-genetic background, and tCycIF data enables the exploration of spatial organization of cells in TME. Spatial profiling was performed on available tissue samples. Description of the cohort and available data is shown in Figure 1.

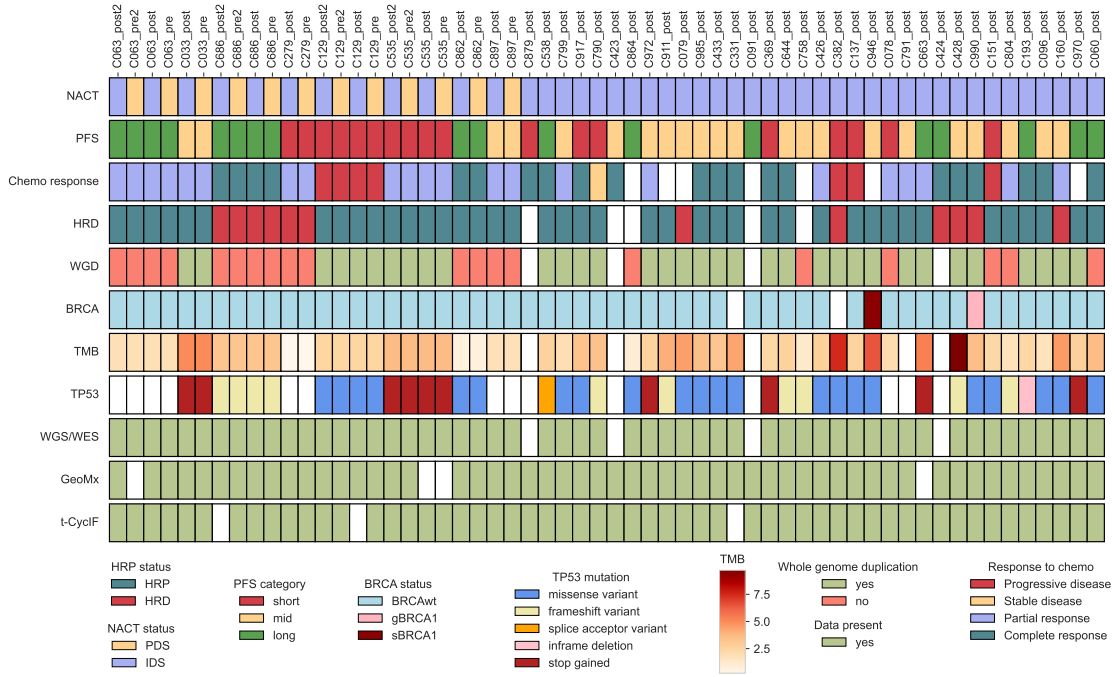


Figure 1: Description of the cohort and available data.

After phenotyping the tCycIF data is processed through *SPACEstat* algorithm that aims to extract features that describe different aspects of the TME. The spatial data obtained from SPACEstat is divided into three levels: single cells, myelonets and lymphonets, and immune communities. Single cell level features describe the organization of single cells in the samples. From single cell data, single cell networks for each cell type are formed, where nodes of the network correspond single cells in sample. From these single cell networks the connected components are extracted. These connected components are referred to as myelonets and lymphonets, depending on whether they are formed by myeloid cells or CD8+/CD4+ T cells, respectively. From lympho- and myelonets, a new network is constructed, where each node corresponds to a lympho- or myelonet. From this network, immune communities of multiple cell types are extracted using Leiden community detection algorithm. These immune communities were then assigned into immune community clusters using K-nearest neighbors by their cell composition.

These layers, clinical, genomic, and spatial information, form our multi-omics data layers that are then integrated for therapy outcome prediction.

2.1 Data characteristics

The clinical data contains clinical information about each patients. This data includes age at diagnosis, stage of the disease, whether the sample of the patient was obtained in Primary Debulking Surgery (before chemo) or Interval Debulking Surgery (after chemo), Homologous Recombination Deficiency (HRD) status, Tumor Mutation Burden (TMB), and primary surgery residual which reports if there is tumor left after Debulking Surgery. Genomic data includes information on genetic mutations in the patient, such as chromosome-level weighted Loss Of Heterozygosity (LOH), Whole Genome Duplication presence, Gene Level Copy Number, ploidy, and purity.

Spatial features include some basic variables such as cell type fractions across the whole sample and in stromal and tumor compartments. Different density metrics for single cell variables are obtained from SPACEstat algorithm. These include Cluster Density, Nearest Neighbor Distance (NND), and Gini Density. In addition, the

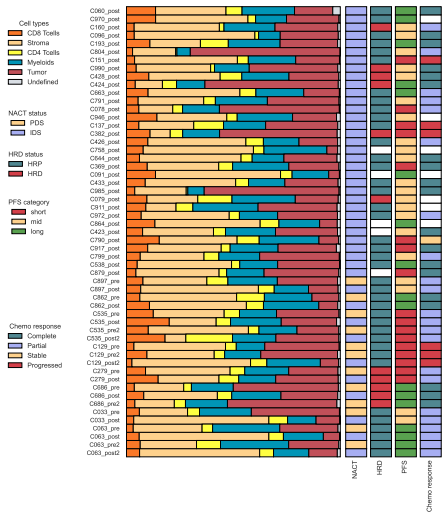
distance to tumor-stroma-interface (TSI) is reported from both inside and outside tumor. Immune communities better describe possible immune cell interactions in the TME.

The primary outcome evaluated in this project was progression-free survival (PFS), measured in days. PFS is defined as the time between the diagnosis of the patient and the first time the disease is detected to have progressed, which in Oncosys-OVA is determined by the imaging results. The PFS is therefore the days between the date of the diagnosis and the date that routine imaging which shows progression, was performed, or the end of study, if the disease had not progressed. Maximum reported progression-free survival in the dataset was 1145 days and minimum was 191 days. Survival categories were computed by taking the bottom and top 25% survival times of the whole cohort and labeling them to short-survival and long-survival groups, respectively. The remaining 50% of patients were categorized to the intermediate survival group. The threshold for short survival group was around 300 days and for long survival group was around 630 days. Because progression was determined from routine imaging, the recorded progression date represents the date which progression was detected rather than necessarily the exact biological onset of progression. Therefore, the measured PFS may contain a small temporal uncertainty. PFS is the most used parameter for evaluating patient outcomes in the context of ovarian cancer, in addition to Overall Survival.

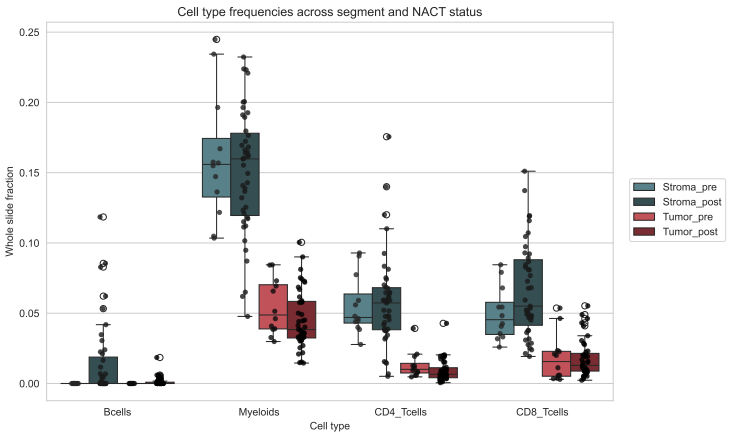
2.2 Exploratory data analysis

Exploratory data analysis was performed for each single-omics dataset separately to find features associated with long survival and better treatment response, and possibly underlying relationships between features. The main goal has been to understand the distributions of features and variability across the cohort, in addition to finding if there are some obvious relationships or patterns with the outcome label.

Figure 2a shows the cell type distributions in samples with some annotated clinical information. The cell type composition of TME across samples is highly variable, which is expected. When looking at the distribution of immune cells in the stromal and tumor compartments in chemo-naïve and post-chemo samples in figure 2b, we can see that the majority of immune cells are located in the stromal regions, and that myeloids are the most abundant immune cells. There seems to be no significant difference between chemo-naive and post-chemo samples.



(a) Cell type proportions across samples from tCycIF data.



(b) Cell type distribution to stroma and tumor regions in chemo-naive and post-chemo samples.

Figure 2: Overview of cell type composition in the tCycIF data.

Gini density describes how evenly different cells are distributed across the sample. Gini coefficient of zero implies that all cells are uniformly distributed across the sample, where as one implies that cells of same cell type are segregated in the sample. This variable tells us more detailed information about the spatial organization of immune cells compared to plain cell fractions. When gini coefficients of immune cells are compared based on the primary treatment response in Figure 3, we can see that the Gini coefficient of CD4+ T cells was higher in progressive disease samples, compared to partial and complete response samples.

Finally, Figure 4 shows the proportions of immune community clusters across the samples. The samples are made up of different proportions immune community clusters, however there is some clear clustering among

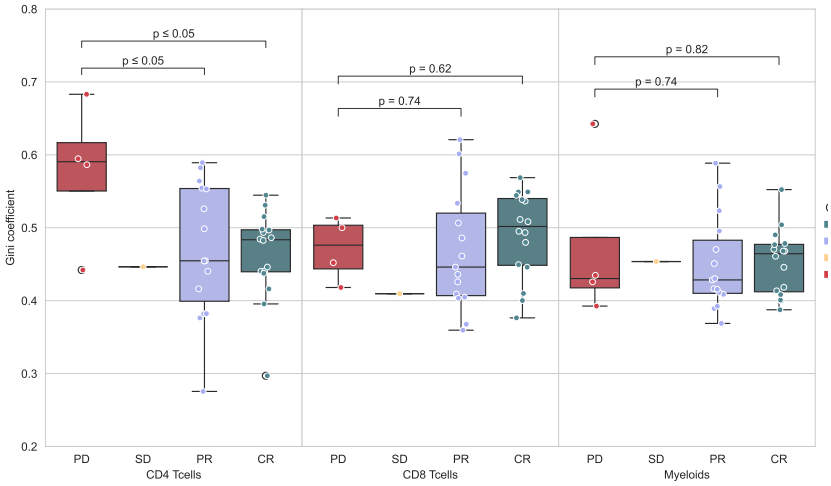


Figure 3: Gini densities of different cell types across response categories to chemo therapy

the samples based on the cluster composition. It remains unclear whether this phenomenon is associated with clinical variables.

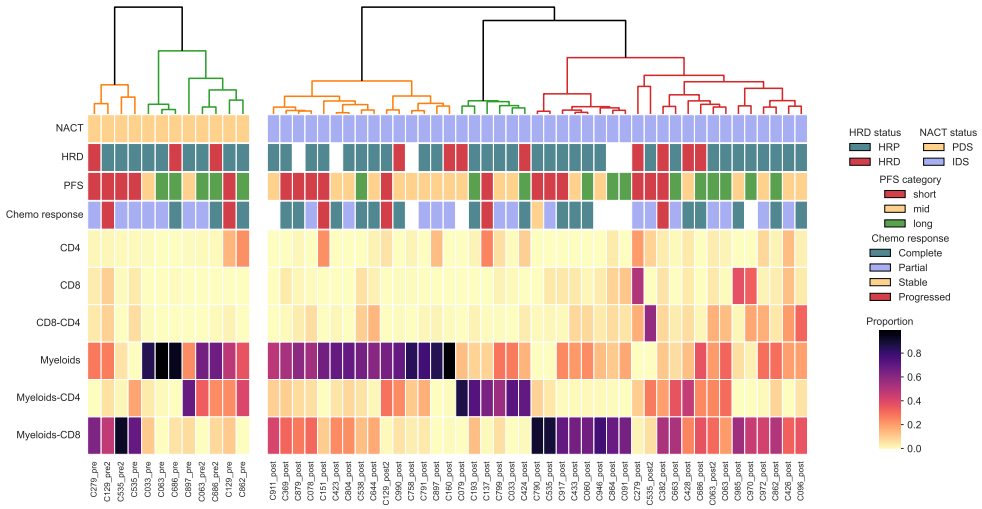


Figure 4: Cluster proportions across chemo-naive and post-chemo samples, ordered by hierarchical clustering.

From the explorative data analysis, we can see that there is lots of heterogeneity among the samples and wide range of different information to use as features. It needs to be noted that the explorative data analysis was conducted over the whole cohort and may therefore not be seen in the further modeling part.

2.3 Data pre-processing

Data provided for this project was mostly ready to use. Some small pre-processing steps were still required to use it for modeling. For binary variables in clinical and genomic datasets, 'yes' and 'no' values were converted to 1 and 0 respectively. Categorical variables such as primary response to chemo therapy and stage of disease was encoded to binary variables for the exploratory data analysis. These categorical variables were not used as candidate features in the modeling. From genomic data, the Gene Level Copy Number values were categorized to Amplification, no mutation, and Deep deletion. For spatial data, different cell type fractions and immune community proportions and immune infiltration indices of each sample were calculated.

In the modeling phase, only post-chemo samples were selected and samples with missing values in the features required were excluded from analysis. This allowed a total of 36 samples from 34 patients to be used for the modeling. Continuous features were standardized using the mean and standard deviation calculated from the training data within each cross-validation fold. The same transformation was then applied to the corresponding test set.

3 Methods

This section describes different methods utilized in the project for both feature selection and modeling.

The aim of this project is to maintain interpretability and reproducibility. Thus, feature selection methods and chosen models need to be explainable. Feature selection enables the identification of biological features associated with clinical outcomes while reducing dimensionality and improving model performance. Although feature extraction methods can improve predictive performance further, feature selection was prioritized in this project to preserve biological interpretability. For outcome prediction, simple machine learning models and survival analysis are preferred to ensure the explainability of the results. For this project, the following python libraries were used: scikit-learn, lifelines, and feature_engine.

Because several patients contributed multiple samples, model performance was estimated with nested patient-grouped cross-validation. In the outer loop, samples were split into five folds with grouped k-fold cross-validation, using the patient as the grouping variable, and shuffled with a fixed seed. Grouped k-fold cross-validation ensured that all samples of a patient were assigned to the same fold to prevent data leakage between training and test sets. One fold of five was used as test set. All data-driven steps were performed using the outer training fold only. The only exception was the preliminary feature selection, which is discussed in Section 5.1.

3.1 Feature selection

The data is high dimensional low sample size (HDLSS), which is a common challenge in multi-omics data analysis. Feature selection is therefore important to reduce the risk of overfitting by excluding noisy and redundant features. A hybrid approach with three stages was used.

First, preliminary selection was performed separately for each single-omics dataset to remove redundant and uninformative variables. This selection was based on domain knowledge and inspection of Spearman correlation heatmaps between features and with the outcome label. Due to the large number of candidate features in the genomic dataset, mRMR with FCD-scheme (F-statistic relevance and correlation-based redundancy) feature selection algorithm (described below) was used to select five most relevant genomic features. After preliminary feature selection, all single-omics datasets were concatenated into a multi-omics dataset.

Second, within each outer training fold, mRMR feature selection was performed. The outer training fold was further divided into five patient-grouped inner folds and mRMR was performed on each of the five inner training sets to select ten most relevant features. Features selected in at least three of the five subsets were considered stable and formed the feature set of that outer fold. Examining the stability of feature selection reduces the dependence on single split of the training data. Because selection was repeated in each outer fold, each fold had their own feature sets and the outer test fold was never used for feature selection. After feature selection, continuous features were standardized using the mean and standard deviation of the outer training fold. Binary features, proportions, and fractions were not standardized.

Third, embedded feature selection with LASSO was performed during the training of linear regression model. The regularization strength α was tuned by an inner grouped 3-fold cross validation on the outer training fold and features with non-zero coefficients were considered selected. For survival models, the mRMR-selected features were used as input.

Minimum Redundancy, Maximum Relevance (mRMR) feature selection algorithm (mRMR) was originally developed for feature selection of gene expression data, but has since been used in other applications as well [1, 7]. In a study focusing on feature selection strategies for multi-omics data [4], mRMR and permutation importance of random forests (RF-VI) were found to work best with few selected features (nvar=10), which makes them good candidates for this application. mRMR iteratively selects the most relevant features with respect to the outcome while minimizing the redundancy with the already selected features. In this project, the feature-engine implementation of mRMR with MID-scheme was applied for selecting the feature set for outer training folds. In the MID-scheme the features were selected based on the difference between relevance and redundancy, which both were calculated based on mutual information. MID-scheme was chosen as it measures non-linear association between features and handles both continuous and discrete variables.

3.2 Model selection

Two different models were chosen to model the patient outcome based on the samples in this project: Linear Regression and Accelerated Failure Time (AFT) survival models. These simple machine learning and survival models were chosen to predict the PFS based on multi-omics data.

3.2.1 Linear regression

Linear regression was included as a simple reference model. Although PFS is a censored time-to-event outcome and linear regression does not explicitly account for censoring, linear regression provides a reference point for

evaluating the performance of more appropriate survival models. The relatively small number of censored observations, 2 of the 36 samples, in the cohort allowed us to use it as an exploratory comparison with other models. Before training linear regression, embedded feature selection using Least Absolute Shrinkage and Selection Operator (LASSO) was performed. The regularization parameter was tuned by an inner 3-fold grouped cross-validation, with 100 values of α between 10^{-4} and 10^{2} and mean absolute error as criterion.

An ordinary linear regression was then fitted using the features with non-zero LASSO coefficients. Mean Absolute Error (MAE) and R^2 score was used as evaluation metrics for Linear regressor. MAE was selected as a evaluation metric as it measures how close the model's predicted survival times are to the observed PFS values. MAE was calculated in two ways: based on all patients and based on those patients whose disease had progressed during the study. R^2 tells how well the model explains the variance of target variable.

3.2.2 Accelerated Failure Time survival models

Second, survival analysis is performed using AFT models with three different distributions, Weibull, LogNormal, and LogLogistic, using lifelines and the mRMR-selected features. Survival models allow the modeling of censored data, which makes them suitable for this application. Progression was used as the event and PFS in days as the duration. AFT models were evaluated based on MAE and R^2 score by considering the predicted median survival of patient, that is the time at which the predicted survival probability is 50%, to enable the comparison with linear regressor. MAE was calculated in two ways, as for linear regression. The primary metric used to evaluate the performance of survival models was concordance index. Concordance index, also known as C-index, is an evaluation metric of survival models and tells, whether the model can correctly rank patient by their survival. 0.5 corresponds to random ranking and 1 perfect ranking.

4 Results

This section presents the results of both feature selection and clinical outcome prediction. Feature selection was performed using domain-knowledge and correlation based filtering, and mRMR algorithm, followed by model-specific embedded feature selection where applicable. Two modeling approaches were evaluated: Linear Regression and AFT survival models.

4.1 Feature selection

In the preliminary feature filtering, redundant and uninformative features were removed from single-omics datasets based on domain-knowledge and inspection of correlation heatmaps. Second, mRMR was performed for the training sets of each cross-validation fold of combined dataset. Third, LASSO was applied during the training of linear regression model to obtain a sparse feature set for linear regressor.

Four selected features from clinical data included primary surgery residual, HRD, age at diagnosis, and TMB. In spatial datasets raw cell counts and features describing B cells and undefined cells were excluded, in addition to whole-slide values being preferred over compartment specific values. Gini density was selected to describe the spatial organization of myelo- and lymphonets and distance to tumor-stroma interface was selected from within the tumor. For immune community clusters, average size of communities was preferred over the biggest community size in cluster. The selection of spatial features was based on domain-knowledge and redundancy between features. mRMR feature selection algorithm was applied for the whole genomic dataset to retrieve five most relevant genomic features. These included fraction of the genome, where the major allele copy number is greater than or equal to 2 ('frac_major_ge2'), presence of whole genome duplication, copy numbers of both PTEN and KRAS, and weighted loss of heterozygosity of chromosome 10.

Figure 5 shows the frequencies of how many outerlinear folds each feature was selected as stable by mRMR. Each outer fold used its own stable feature set, which contained on average 8.8 features. Nine features were stable in at least for three of the five outer folds. These included three genomic features (copy numbers of PTEN and KRAS, and weighted loss of heterozygosity in chromosome 10) and six spatial features (Whole slide proportion of cells belonging to myeloid dominated cluster and CD4+ T cell dominated cluster, average size of communities in Myeloid and CD4+ T cells dominated cluster and in CD8+ T cell dominated cluster, fraction of myeloid cells in whole slide, and CD8+ T cells' gini density), four of which described the immune community clusters.

The feature selection stability of LASSO over outer cross-validation folds is displayed in Figure 6. No feature had non-zero LASSO coefficient for all five folds. On average, LASSO retained 4.4 features per fold. LASSO was applied to the mRMR-selected features. Genomic features were the most stable, except for 'frac_major_ge2', which was used in only one of five folds. Copy number of PTEN and KRAS were the most consistently selected features, as they were selected four times, and weighted loss of heterozygosity of chromosome 10 was selected in three of cross-validation folds. All spatial features that were considered as stable in the mRMR were only used in at most two outer folds.

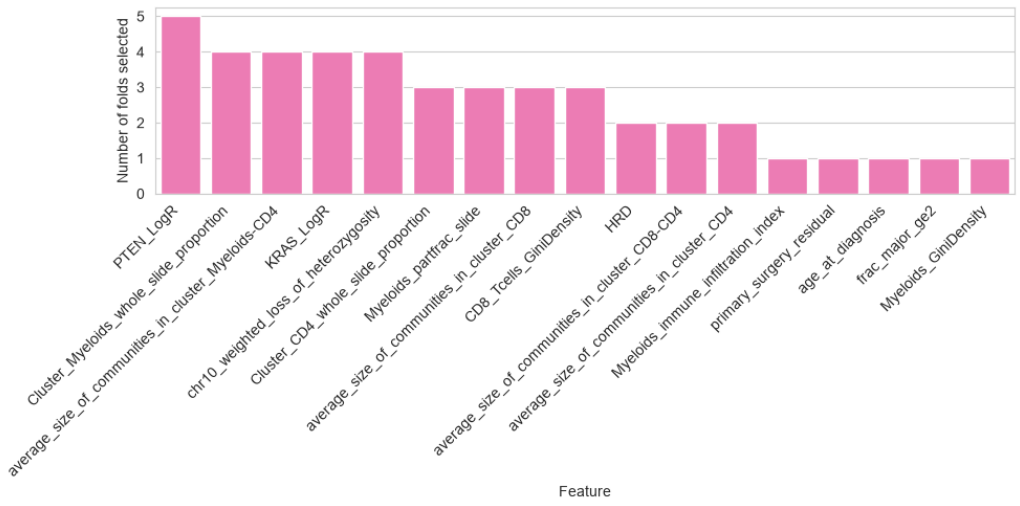


Figure 5: Results of mRMR feature selection stability over outer cross-validation folds.

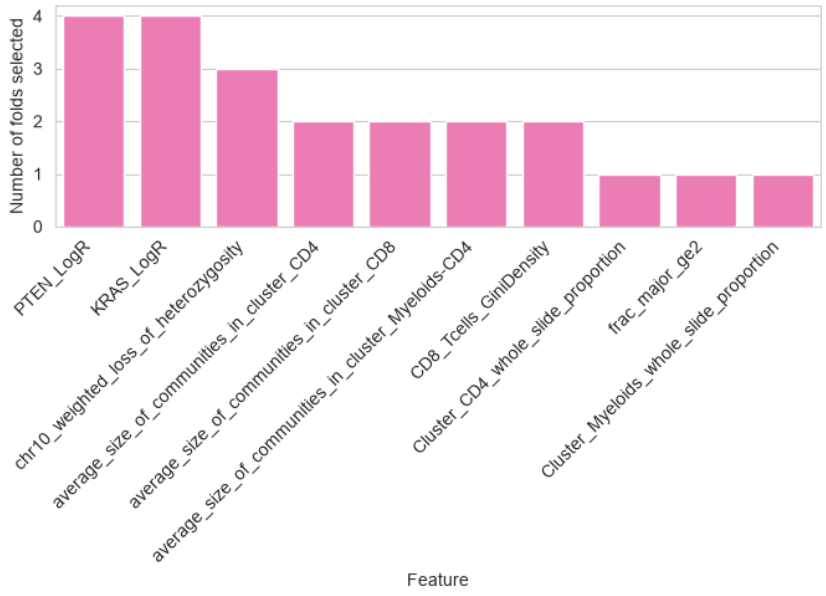


Figure 6: Results of non-zero features of LASSO over outer cross-validation folds.

4.2 Model performance

Model performance was estimated with patient-grouped cross-validation, where both models used identical folds for training and evaluation. Predictive performance of models is reported as the mean, standard deviation, minimum, and maximum over the five outer test folds.

4.2.1 Linear regression

The performance of linear regression was evaluated using MAE and R^2 of cross-validation folds. The results are displayed in Table 1. On average, the predicted survival days differed from the observed PFS days by ≈ 174 days calculated based on all patients, and ≈ 171 days for those patients were reported for having progressive disease. This means that the prediction is off by almost six months from the observed survival. The mean R^2 was -0.46, meaning that on average, the predictions were less accurate than simply predicting the mean PFS of the test fold.

When looking at the standard deviation of evaluation metrics, one can see that the model performance varied drastically between cross-validation folds. The standard deviation of MAE for all patients was ≈ 55 days and for progressed patients ≈ 50 days. In the best fold, model was mistaken by 96 days (approximately three months) for both all patients and progressed patients. In the worst fold, MAE for all patients was 249 days. Same pattern can be seen when inspecting the standard deviation of R^2. In the best fold, R^2 reached 0.60, but in the worst it was -2.45.

Metric	Mean	SD	Min	Max
MAE (days)	173.64	54.57	95.60	249.41
MAE, progressed only (days)	170.98	49.80	95.60	235.16
R^2	-0.46	1.17	-2.45	0.60

Table 1: Linear regression performance over the five outer folds.

4.2.2 Accelerated Failure Time survival models

The results of AFT survival models are displayed in Table 2. For all survival model distributions, the MAE of all patients and only progressed patient were close to each other. Best MAE, averaged over cross-validation folds was for AFT model with LogLogistic distribution. The mean of both MAE measures were 153 days, approximately five months. AFT with Weibull distribution differed by approximately 161 days on average, and AFT with LogNormal distribution was off by approximately 165 days on average, for both all patients and progressed only patients. AFT survival model with LogLogistic distribution had R^2 of -0.50, whereas the rest of the AFT models had R^2 of approximately -1.7. All models had modest mean concordance indices; 0.67 for LogNormal and 0.70 for LogLogistic and Weibull.

Model	Metric	Mean	SD	Min	Max
LogLogistic	MAE (days)	152.79	73.59	63.60	235.36
	MAE, progressed only (days)	152.94	70.66	69.02	230.68
	R^2	-0.50	1.81	-3.65	0.77
	C-index	0.70	0.13	0.53	0.88
LogNormal	MAE (days)	165.59	73.89	76.50	240.08
	MAE, progressed only (days)	164.87	71.11	79.97	233.00
	R^2	-1.67	4.28	-9.30	0.66
	C-index	0.67	0.10	0.53	0.76
Weibull	MAE (days)	160.74	87.79	39.66	249.19
	MAE, progressed only (days)	160.10	84.11	45.14	249.19
	R^2	-1.68	4.58	-9.83	0.90
	C-index	0.70	0.18	0.53	1.00

Table 2: AFT model performance over the five outer folds.

Same observation can be seen in the evaluation metrics of survival models as for linear regression. The performance of models vary drastically between cross-validation folds. For all AFT models, the standard deviation of both MAE exceeded 70 days. For LogNormal and Weibull distributions, R^2 varied drastically between folds $(4.28, 4.58)$, while the variance with LogLogistic distribution was milder (1.81), although still high.

Compared with linear regression, Table 1, all AFT models had lower mean MAE, the LogLogistic model by approximately 21 days. In terms of R^2, only LogLogistic model (-0.5) was comparable to linear regression (-0.46), although both results are poor.

5 Discussion

The aim of this project was to identify biological features associated with progression-free survival in HGSC and to evaluate whether simple, interpretable models can predict the patient survival from clinical, genomic, and spatial data. Models were evaluated with nested, patient-grouped five-fold cross-validation with identical outer folds for all models. The predictive performance of all models was modest.

Linear regression explained none of the variance in PFS with mean R^2 value of -0.46. The mean absolute error for all patients of 174 days, almost six months, is also large compared to the observed PFS in the cohort, which ranged from 191 to 1145 days with median of 467 days. Calculating the MAE based on uncensored patients did not substantially better the result. The AFT models reached a mean C-indices of $0.67 - 0.70$, indicating that they ranked patients by PFS better than random. All three distributions performed similarly. The predicted survival times were also assessed based on MAE. All three survival models were moderately more accurate than linear regression, although based on R^2 they were not. The AFT models performed the best with a mean C-index of approximately 0.8 and all three distributions performed similarly. As the C-index evaluates only the ordering of patients, the predicted survival times were also assessed. Based on MAE and R^2, the survival models performed slightly better than linear regression.

A variety of factors contribute to the large prediction errors and the variation between folds. First, each test

fold contained only six to eight samples, which causes a few poorly predicted patients to have a large effect on fold-level performance metrics, both MAE and C-index. Second, the progression is detected at routine imaging, which enables that the recorded progression date may lag behind actual progression by up to one imaging interval. Third, the number of features was large relative to the number of samples. With approximately 28-30 training samples and up to 11 features per fold, the risk of overfitting increases.

Copy number of PTEN and KRAS, and weighted loss of heterozygosity of chromosome 10 were the most consistently selected features by both mRMR and LASSO. Of the spatial features, average size of immune communities in CD8+ T cell dominated cluster, and Myeloid and CD4+ T cell dominated cluster, and gini density of CD8+ T cells were selected fairly consistently. The average size of immune communities in CD4+ T cell dominated cluster was considered as stable feature twice but both times LASSO also included it as a feature to use for modeling.

The mix of both genomic and spatial features among the most stable features suggests that both layers carry information about PFS. However, no single-omics models were compared and the genomic features were selected by using the whole cohort. Therefore, these features should be treated as hypothesis-generating candidates and not as validated predictors.

5.1 Limitations

This project has some limitations that must be addressed. One limitation of this project is the information leakage in the preliminary feature selection phase, which means that the test folds were not completely independent from the feature selection. The genomic features were selected with mRMR using PFS on the whole post-chemo cohort.

Another significant limitation of this project was the small sample size. Only post-chemo samples were used for modeling, leaving 36 samples from 34 patients. Consequently, each test fold contained only six to eight samples, resulting in large standard deviation of evaluation metrics.

5.2 Future work

Several extensions would strengthen the analysis and address the limitations described above. First, the outcome-based preliminary feature selection should be performed based on the training set of outer cross-validation fold to minimize the information leakage. Second, all models could be compared against models trained on each omics layer separately to determine whether the combination of clinical, genomic, and spatial data improves prediction. Third, more accurate, cencoring aware performance measures would allow a more appropriate assesment of the survival models than MAE and R^2. Finally, a larger, external cohort is needed to validate the candidate features before conclusions on their prognostic relevance can be drawn.

6 Conclusion

The aim of this project was to identify a set of features associated with progression free survival in HGSC and examine how it can be predicted from clinical, genomic, and spatial omics data, using simple, interpretable models evaluated by nested, patient-grouped cross-validation. A hybrid feature selection approach combining domain-knowledge, correlation-based filtering, mRMR and LASSO reduced the high-dimensional data to a small set of genomic and spatial features. The AFT models ranked patients by PFS better than random, with mean C-indices of $0.67 - 0.70$, however, none of the models predicted PFS in days accurately and the performance between folds varied considerably. Copy number of PTEN and KRAS, weighted loss of heterozygosity of chromosome 10, and spatial features describing myelo- and lymphonet dominant immune communities were the most consistently selected features. The results suggest that genomic background and spatial organization of the TME carry information about disease progression of HGSC, as previous studies have shown [3]. Due to the small cohort and partial information leakage in the feature selection, these findings are hypothesis-generating and require validation in a larger, independent cohort.

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